AWS Lab: Multi-Region Resource Discovery & CLI Region Targeting

Duration : 30 Min.

Scenario

An application developer cannot see their newly created EC2 instance in the AWS Management Console and assumes provisioning failed. In reality, the console defaulted to us-east-1 while the instance was created in ap-south-1. The engineer must use the AWS CLI to query instances across regions, locate the missing server, and update the developer's default AWS CLI profile to prevent duplicate provisioning.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon Web Services (AWS CLI) - **Target Region:** `ap-south-1` (Asia Pacific - Mumbai) - **Target Availability Zone:** `ap-south-1a` - **Instance Name Tag:** `web-app-prod`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Run aws ec2 describe-instances across regions to locate instance 'web-app-prod'. 2. Identify the target Region (ap-south-1) and Availability Zone (ap-south-1a). 3. Configure default region targeting in AWS CLI profile that configures AWS_DEFAULT_REGION properly.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Instance discovery: correctly query multiple regions to locate the hidden instance	4 Marks
TC2	Topology identification: identify the exact Region and AZ housing the resource	3 Marks
TC3	Configuration update: set_default_region.sh correctly sets the environment default	3 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.